

[standalone build] Skew-brace decomposition obstruction chapter

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Chapter 1

The decomposition obstruction of a skew brace: a bilinear class, and its residue-dependent solvability

A skew brace is the algebraic shadow of a set-theoretic solution of the Yang–Baxter equation, and the recurring engineering question about it is the same one that runs through this whole dossier: *when does a composite come apart into its pieces?* Concretely, given a skew brace G and an ideal $D \trianglelefteq G$, when is G an internal bicrossed (Zappa–Szép) product $(G/D) \bowtie D$? This chapter answers three questions about the obstruction to that decomposition. It *names* the obstruction (a class in Rathee–Yadav skew-brace cohomology, [PEER-REVIEWED]); it *measures* the obstruction (a genuinely *bilinear* class, strictly finer than the two underlying group Schreier classes, [PROVED]); and it asks how the obstruction relates to the one datum most easily confused with it — the internal relabelling residue $[\nu]$. The honest answer to the third question is the point of this chapter, and it is not the one an earlier draft gave.

A correction on the record. A 2026-09-26 draft of this chapter claimed a *third theorem*: that the residue $[\nu]$ and the obstruction $[\Omega]$ are *independent coordinates*, so that the moduli of extensions factors as a product $R \times \Omega^0$. That claim is **false**. It was refuted by Rick (referee report, 2026-09-26) and reproduced from scratch (2026-09-29): the residue $[\nu]$, the circle action σ , and the additive class $[\beta]$ (hence $[\Omega]$) are genuinely *coupled*. The false argument reduced the existence question to “read $\circ$ off from $+$ and λ ” — which presupposes the very brace whose existence was in question. What replaces the product theorem is sharper and true: a *residue-dependent* solvability obstruction o_ν , a homomorphism whose kernel is exactly the set of additive classes that extend to a full brace with the prescribed residue (§1.5). The coupling, not the independence, is the content.

The result answers, in group-theoretic language, Ferri’s Remark 3.20 on when a quiver skew brace splits off an ideal [1], and it supplies the skew-brace / Yang–Baxter instance of the dossier’s decomposition programme. A word of warning before we begin, because the two are so easily conflated: the *existence* axis — “does G admit a Zappa–Szép factorisation at all” — is the subject of Chapter ?? and is governed by the group-cohomological class $[\omega]$. The *decomposition* axis of this chapter — “does G factor *through a prescribed ideal* D ” — is strictly finer, and its obstruction $[\Omega]$ lives in a two-operation cohomology that no single group class can see. Keeping the two apart is one point; keeping the obstruction apart from the residue-*solvability* question is the other.

1.1 The three theorems

Throughout, $(G, +, \circ)$ is a skew left brace: $(G, +)$ and $(G, \circ)$ are groups sharing the identity 0, related by $a \circ (b + c) = (a \circ b) - a + (a \circ c)$, and we write $\lambda_a(b) := -a + (a \circ b)$, so $a \circ b = a + \lambda_a(b)$ and $\lambda: (G, \circ) \rightarrow \text{Aut}(G, +)$ is a homomorphism. An *ideal* $D \trianglelefteq G$ is a normal subgroup for both operations with $\lambda_a(D) = D$; it yields a short exact sequence of skew braces

$$0 \longrightarrow D \longrightarrow G \xrightarrow{\pi} H := G/D \longrightarrow 0.$$

We assume $(D, +)$ and $(D, \circ)$ are both abelian — the setting in which the extension is classified by a second cohomology — and, where a citation to Rathee–Yadav’s theory is invoked directly, the standing hypothesis (SH) that D is a *trivial* brace ($a \circ b = a + b$ on D). A *decomposition of G through D* is an isomorphism $G \cong (G/D) \bowtie D$ onto an internal bicrossed product; equivalently (see §1.3) a sub-skew-brace complement to D .

Fix a normalised section $s: H \rightarrow G$ (so $s(0) = 0$). It determines the *good triplet*

$$\nu_h := \lambda_{s(h)}|_D, \quad \mu_h(x) := -s(h) + x + s(h), \quad \sigma_h(x) := s(h)^{-1} \circ x \circ s(h),$$

of maps $D \rightarrow D$, together with the additive and circle factor sets

$$\beta(h_1, h_2) := -s(h_1 + h_2) + s(h_1) + s(h_2) \in D, \quad \bar{\tau}(h_1, h_2) := s(h_1 \circ h_2)^{-1} \circ s(h_1) \circ s(h_2) \in D.$$

The three theorems of the chapter, stated informally now and precisely in the body:

Theorem 1.1 (Identification, [PEER-REVIEWED]; §1.3). *The obstruction to decomposing G through D is a single class $[\Omega] \in H_{\text{sb}}^2(H, D)$ in Rathee–Yadav skew-brace cohomology, and $[\Omega] = 0$ if and only if $G \cong (G/D) \bowtie D$, i.e. D has a sub-skew-brace complement. No coprimality hypothesis is needed.*

Theorem 1.2 (Bilinearity, [PROVED]; §1.4). *$[\Omega]$ is a genuinely bilinear class: the two forgetful maps to the additive and circle group Schreier classes, $\varphi_+([\Omega]) = [\beta]$ and $\varphi_\circ([\Omega]) = [\bar{\tau}]$, can both vanish while $[\Omega] \neq 0$. Vanishing of both single-operation obstructions does not imply decomposition. The obstruction is base-localised: it is decided on a single star of the ambient groupoid, and closure there propagates to all stars.*

Theorem 1.3 (Residue-dependent solvability, [PROVED]; §1.5). *The internal relabelling residue $[\nu]$ and the obstruction are not independent. Under (SH) there is a residue-dependent solvability obstruction — a group homomorphism*

$$o_\nu: H_+^2 := H_{\text{gp}}^2((H, +), D; \mu) \longrightarrow C^3/A(Z_\circ^2)$$

killing additive coboundaries — such that an additive class $[\beta]$ extends to a full skew-brace cocycle inducing the fixed triplet if and only if $o_\nu([\beta]) = 0$; equivalently $\text{im } \varphi_+ = \ker o_\nu$. Because o_ν depends on the residue ν , the set of realisable additive classes moves with the residue. Explicitly: at $|G| = 8$, residue $\nu = 1$ forces $[\beta] = 0$; at $|G| = 16$ the residue, circle action, and additive class are locked by a three-way relation $4b \equiv 2(\nu\sigma - 1) \pmod{8}$; and for a non-trivial kernel o_ν becomes affine, so $\text{im } \varphi_+$ is a torsor that can miss the origin.

The method is uniform and worth stating in one breath. Because $a \circ b = a + \lambda_a(b)$, the circle factor set $\bar{\tau}$ is not a free coordinate: given a genuine brace it is *determined* by $(G, +)$ and λ . But this determination presumes a brace already exists — it cannot be used to *manufacture* one.

The real question, “does an extension with a prescribed residue exist at all?”, is the solvability of an inhomogeneous linear system, and its obstruction is the class o_ν . The residue $[\nu]$ is the $D \rightarrow D$ block of λ ; the obstruction $[\Omega]$ is the off-diagonal, section-varying part. The block-triangular structure of the brace axioms cuts out the *admissible* residues by an action-only equation (§1.2); but it does *not* decouple *solvability* — the inhomogeneous term of the coupling equation carries the residue, and that is where o_ν lives.

Outline. §1.2 records the factor-set calculus and the coupling equation, and flags exactly where the residue enters. §1.3 proves Theorem 1.1 (identification). §1.4 proves Theorem 1.2 (bilinearity and base-localisation). §1.5 constructs the solvability obstruction o_ν and proves $\text{im } \varphi_+ = \ker o_\nu$ (Theorem 1.3), carrying along the residue-invariance lemma and the sharp negative that orthogonality fails already at the cochain level. §1.6 makes the residue-dependence explicit at $|G| = 8$ and $|G| = 16$ and states the torsor phenomenon. §1.7 is the honest ledger — including the refuted claim on the record — and the relation to Ferri’s Remark 3.20.

1.2 Setup: factor sets and the coupling equation

With $(G, +)$ fixed, a skew-brace structure making D an ideal is *exactly* a homomorphism $\lambda: G \rightarrow \text{Aut}(G, +)$ satisfying $\lambda_{a \circ b} = \lambda_a \lambda_b$ (with $a \circ b := a + \lambda_a(b)$) and leaving D invariant. This is a genuine *existence* question: whether such a λ exists, with a prescribed residue, is not settled by “reading $\circ$ off from $+$ ”. Given a brace, the circle factor set is read off from $\circ$; but the equations below are what decide whether the brace is there to begin with.

Definition 1.4 (Inner group and residue). The *inner group* is $L := \{\lambda_e|_D : e \in D\} = \text{im}(\lambda|_D) \leq \text{Aut}(D, +)$, with $L \cong (D, \circ)/\ker(\lambda|_D)$; write $D_0 := \ker(\lambda|_D)$, so D is a trivial brace iff $L = \{\text{id}\}$ iff $D_0 = D$. The *residue* of the section datum $\nu_h = \lambda_{s(h)}|_D$ is the coset $[\nu_h] := \nu_h L \in \text{Aut}(D, +)/L$.

In coordinates the brace axioms are equivalent to the following system (the generalised Rathee–Yadav conditions; the additive-difference circle factor set is written $T(h, k) := s(h) \circ s(k) - s(h \circ_H k) \in D$):

$$(a) \quad \mu_{h_1} \beta(h_2, h_3) + \beta(h_1, h_2 + h_3) = \beta(h_1, h_2) + \beta(h_1 + h_2, h_3), \quad (a)$$

$$(b) \quad T \text{ is a circle 2-cocycle (the } \circ\text{-associativity condition),} \quad (b)$$

$$(c) \quad \mu_h \nu_h \beta(k, l) + T(h, k +_H l) = T(h, k) - \mu_{h \circ k} \mu_h^{-1} \beta(h, -h) + \beta(h \circ k, -h) \\ + \mu_{(h \circ k) - h} T(h, l) + \beta((h \circ k) - h, h \circ l), \quad (c)$$

$$\text{(coupling-1)} \quad \nu_h \mu_k \nu_h^{-1} = \mu_{\lambda_h^H(k)}. \quad (*)$$

Here $\lambda^H: (H, \circ_H) \rightarrow \text{Aut}(H, +)$ is the induced H -brace map. Two structural facts about this system carry the chapter, and we must state them with care, because an earlier draft over-read them:

- Equation (coupling-1) constrains the residue using only ν, μ, λ^H — it mentions neither T nor β . This is the *admissibility* of a residue: a genuinely block-triangular, obstruction-free condition. It determines *which* residues can occur.
- The residue ν enters (c) through the inhomogeneous term $\mu_h \nu_h \beta(k, l)$. In the trivial-kernel (SH) regime this is its *only* appearance in (a)–(c); but for a non-trivial kernel ν re-enters through λ_d acting on the additive complement (the $\nu \sigma(d)$ term of the complement cochain), so it appears *twice*.

The crucial correction to the earlier draft is this. That the *homogeneous* part of (c) is ν -free (it is: see §1.5) does *not* make the *inhomogeneous* solvability ν -free. Admissibility of a residue is block-triangular; *solvability with that residue* is not. The inhomogeneous term $\mu_h \nu_h \beta$ is exactly what the solvability obstruction o_ν measures, and it carries the residue explicitly. Block-triangularity buys admissibility, and nothing more.

1.3 The obstruction is a skew-brace cohomology class

We first pin $[\Omega]$ down as a cohomology class and record what its vanishing means. All of this section is [PEER-REVIEWED] (referee: Rick); the cohomology machinery is Rathee–Yadav’s [2], invoked, not re-derived.

Under (SH) the pair $(\beta, \bar{\tau})$, subject to (a), (b), (c), represents a class

$$[\Omega] := [(\beta, \bar{\tau})] \in H_{\text{Sb}}^2(H, D) := Z_{\text{Sb}}^2(H, D)/B_{\text{Sb}}^2(H, D),$$

where Z_{Sb}^2 is the set of pairs satisfying the coupling (c) together with $\beta \in Z_{\text{Gp}}^2((H, +); \mu)$ and $\bar{\tau} \in Z_{\text{Gp}}^2((H, \circ); \sigma)$, and B_{Sb}^2 is generated by the section changes below. This is exactly the group that classifies skew-brace extensions inducing the fixed triplet (ν, μ, σ) . The circle datum $\bar{\tau}$ and the additive-difference factor set T of §1.2 are one datum in two coordinate systems: $\bar{\tau}$ is the circle-multiplicative form and $T = \nu_{\text{hok}}(\bar{\tau})$ its additive image. We use T in the cross-equation (c) — where it must interact additively with β — and $\bar{\tau}$ for the circle cohomology class.

Lemma 1.5 (Additive coboundary; $\beta = d\theta$). *Under a section change $s'(h) = s(h) + \theta(h)$ with $\theta(0) = 0$, the additive factor set shifts by an additive coboundary and nothing else:*

$$(\beta' - \beta)(h_1, h_2) = \mu_{h_2}(\theta(h_1)) - \theta(h_1 + h_2) + \theta(h_2) = (d\theta)(h_1, h_2),$$

independent of $\bar{\tau}$, ν and σ .

This is the $\beta = d\theta$ convention of [2]; we adopt it verbatim. The lemma says the additive datum β is exactly the group 2-cocycle of $0 \rightarrow (D, +) \rightarrow (G, +) \rightarrow (H, +) \rightarrow 0$ with μ its conjugation action; so the forgetful assignment $\varphi_+ : [\Omega] \mapsto [\beta] \in H_{\text{Gp}}^2((H, +), D; \mu)$ is well defined — the coupling (c) constrains cocycles but not coboundaries.

Theorem 1.6 (Well-definedness of φ_+ , [PEER-REVIEWED]). *φ_+ is a well-defined homomorphism; the coupling (c) does not obstruct it. The circle forgetful map $\varphi_\circ : [\Omega] \mapsto [\bar{\tau}] \in H_{\text{Gp}}^2((H, \circ), D; \sigma)$ is defined symmetrically.*

Theorem 1.7 (Identification, [PEER-REVIEWED]). *Under (SH), for the extension $0 \rightarrow D \rightarrow G \rightarrow H \rightarrow 0$ with good triplet (ν, μ, σ) ,*

$$[\Omega] = 0 \iff \text{the sequence splits by a skew-brace section} \iff G \cong (G/D) \ltimes D \iff D \text{ has a sub-skew-brace complement}$$

No coprimality hypothesis is required (coprimality enters only Rathee–Yadav’s stronger uniqueness theorem). The first equivalence is intrinsic to the Rathee–Yadav classification; the remaining ones are the registry result corrected-equivalence-1-2-3, via Ferri’s Theorem 3.18 [1].

A *sub-skew-brace complement* is a subset $K \subseteq G$ closed under both $+$ and $\circ$, with $K \cap D = \{0\}$ and $|K| |D| = |G|$; the equivalence records that a one-sided (additive-only or circle-only) complement is strictly weaker, which is precisely the content of the next section. The Lean corpus supplies machine-checked `decide` certificates for the descent of φ_+ and for the non-injectivity witness of §1.4 [LEAN-VERIFIED].

1.4 Bilinearity: strictly finer than both Schreier classes

The heart of the chapter’s “measurement” claim is that $[\Omega]$ is not the pair of group classes $(\varphi_+([\Omega]), \varphi_\circ([\Omega]))$; it is genuinely bilinear, seeing a coupling between the two operations that neither operation sees alone. Everything here is [PROVED].

Theorem 1.8 (The product map is not injective, [PROVED]). *Let $\Phi := (\varphi_\circ, \varphi_+): H_{\text{sb}}^2(H, D) \rightarrow H_{\text{gp}}^2((H, \circ), D; \sigma) \times H_{\text{gp}}^2((H, +), D; \mu)$. Then $\ker \Phi \neq 0$: there is a skew brace with $\Phi([\Omega]) = (0, 0)$ yet $[\Omega] \neq 0$.*

The W_4 witness. Take additive group $(G, +) = \mathbb{Z}/2 \times \mathbb{Z}/4$, multiplicative group $(G, \circ) \cong \mathbb{Z}/4 \times \mathbb{Z}/2$, and ideal $D = \{0\} \times \mathbb{Z}/4$ (a trivial-brace ideal), with $H = G/D \cong \mathbb{Z}/2$. Then:

- the additive extension splits — the additive complement $\{(0, 0), (1, 0)\}$ gives $[\beta] = \varphi_+([\Omega]) = 0$;
- the multiplicative extension splits — the multiplicative complement $\{(0, 0), (1, 1)\}$ gives $[\bar{\tau}] = \varphi_\circ([\Omega]) = 0$;
- but these two complements are *distinct subsets*, and no single subset is a complement for both operations: there is no sub-skew-brace complement, so by Theorem 1.7, $[\Omega] \neq 0$.

Hence $\Phi([\Omega]) = (0, 0)$ with $[\Omega] \neq 0$. The obstruction lives in the coupling term of (c); it is *irreducibly bilinear*. $\square$

The witness is exact and small enough to enumerate; a sweep over $\mathbb{Z}/4$, $(\mathbb{Z}/2)^2$, $\mathbb{Z}/6$, $\mathbb{Z}/8$, $\mathbb{Z}/9$, $\mathbb{Z}/2 \times \mathbb{Z}/4$ produces ten W_4 -type witnesses on $\mathbb{Z}/2 \times \mathbb{Z}/4$, and locates a “phantom” family in which 96 of 144 candidate ideals admit an ambient additive complement while 48 do not — the additive picture alone cannot predict decomposition. Formally, the two forgetful functors send $[\Omega]$ to the additive and multiplicative Schreier classes $[\omega_+] \in H^2((G/D)_+, D_+)$ and $[\omega_\circ] \in H^2((G/D)_\circ, D_\circ)$, and $[\Omega]$ may be nonzero while *either or both* vanish. This is why the decomposition axis of this chapter is strictly finer than the existence axis of Chapter ??: the group class $[\omega]$ is one leg of Φ , and Φ has a kernel.

Base-localisation. In the groupoid (quiver skew brace) generalisation of the problem — where G is a groupoid with a per-star additive structure and D a wide normal sub-bundle — the obstruction is not merely finer, it is *localised*. The naive “heap-transport” conjecture (that a decomposition of the vertex/loop skew brace L_ζ suffices for a decomposition of G) is *false*:

Theorem 1.9 (Vertex splitting is necessary but not sufficient, [PROVED]). *There is a connected quiver skew brace on $V \times \text{Coarse}(\{0, 1\})$ with V the Klein four-group and star group $(\mathbb{Z}/2)^3$, and a proper ideal sub-bundle D , such that the vertex skew brace L_ζ decomposes (D_ζ has a sub-skew-brace complement) yet D has no wide sub-quiver-skew-brace complement, so $G \not\cong (G/D) \bowtie D$.*

The correct invariant is a *star complement*: $G \cong (G/D) \bowtie D$ iff there is a multiplicative complement K of D_ζ in L_ζ together with connecting labels (τ_μ) (with $\tau_\zeta = e$) such that $A = \{(\zeta, \mu, k\tau_\mu) : k \in K, \mu \in X\}$ is an additive complement of D_ζ *closed under $+_\zeta$ in the whole base star*. Closure on one base star forces closure on every star, via the additive isomorphisms $\rho_a = a \triangleright (-)$ carrying D_ζ to D_μ ; so the obstruction is base- ζ localised. For the Klein witness all single-operation shadows split — additive extension of the star, additive complement of D_ζ , multiplicative complement $K = \{0, 3\}$, and the reduced base extension — yet no star complement

exists. This is the same both-split-no-common-complement phenomenon as W_4 , now with the extra lesson that testing the loop shadow instead of the base star is precisely the mistake the heap-transport conjecture made. (A brute-force sweep of the Klein family found 504 vertex-splitting-but-not-decomposing configurations and 0 wide complements; the star-transport lemma held in 1008/1008 cases.)

1.5 The residue-dependent solvability obstruction

We now come to the residue $[\nu]$, the datum most easily mistaken for part of $[\Omega]$. The honest statement is that they are *coupled*, and that the coupling is measured by a single homomorphism o_ν . Two facts about the residue survive from the analysis and are worth isolating first, because they are true and were never in question; it is only the *independence* conclusion drawn from them that failed.

Lemma 1.10 (Residue invariance, [PROVED]). *The cosets $[\nu_h] = \nu_h L \in \text{Aut}(D, +)/L$ are section-independent, and $\nu_{h_1}\nu_{h_2} = \nu_{h_1 \circ h_2} \cdot \lambda_{\bar{\tau}(h_1, h_2)}|_D$ with $\lambda_{\bar{\tau}}|_D \in L$. When $L \trianglelefteq \text{Aut}(D, +)$ (automatic if $\text{Aut}(D, +)$ is abelian, e.g. D cyclic), $[\nu]: (H, \circ) \rightarrow \text{Aut}(D, +)/L$ is a homomorphism. As s' ranges over all normalised sections, $(\nu'_h)_{h \neq 0}$ ranges over exactly the torsor $T_\nu := \prod_{h \neq 0} \nu_h L$ under $L^{H \setminus 0}$, with invariant residue $[\nu]$.*

Proposition 1.11 (Orthogonality fails already at the cochain level, [PROVED]). *The gauge action of the section group on the cochain triple $(\nu, \beta, \bar{\tau})$ is not a direct product of a ν -action and an obstruction-action: $\ker(\Psi_{\text{obs}}) + \ker(\Psi_\nu) \neq \Theta$. For the braces **sb#11**, **sb#15** on $\mathbb{Z}/2 \times \mathbb{Z}/4$ with $D = \{0\} \times \mathbb{Z}/4$ ($L = \text{Aut}(D, +) = \mathbb{Z}/2$), any section that flips ν_1 forces $d\theta(1, 1) = (0, 2) \neq 0$: moving the residue drags a nonzero change in the obstruction cochain.*

Proposition 1.11 was, in the refuted draft, the “fence” meant to keep an independence claim from overreaching. In hindsight it was the warning that the claim should not have been made: the residue and the obstruction cochain move *together*, and the only question is whether the coupled motion is a coboundary (invisible to cohomology) or genuine. It is genuine, and o_ν is what detects it.

The coupling operators. Write C^k for the normalised k -cochains $H^k \rightarrow D$. Read the coupling (c) as an affine-linear equation in the multiplicative datum, for fixed β . Define two $\mathbb{Z}$ -linear operators $C^2 \rightarrow C^3$:

$$(P\tau)(h_1, h_2, h_3) := \mu_{\lambda_{h_1}^H(h_3)}(\tau(h_1, h_2)) - \tau(h_1, h_2 + h_3) + \tau(h_1, h_3), \quad (1.1)$$

$$(Q\beta)(h_1, h_2, h_3) := \nu_{h_1}(\beta(h_2, h_3)) + \mu_{h_1 \circ h_3}(\beta(h_1, -h_1)) - \beta(-h_1, h_1 \circ h_3) - \beta(h_1 \circ h_2, \lambda_{h_1}^H(h_3)). \quad (1.2)$$

Both operators depend only on the fixed triplet and on the group structure of H — *no brace is presupposed*, which is exactly the circular step the refuted argument stumbled over. With the twist $T: C^2 \rightarrow C^2$, $(T\bar{\tau})(h_1, h_2) := \nu_{h_1 \circ h_2}(\bar{\tau}(h_1, h_2))$ (so the additive form is $\tau = T\bar{\tau}$), the coupling (c) reads

$$P(T\bar{\tau}) = Q\beta. \quad (1.3)$$

Let $A := P \circ T$ restricted to $Z_\circ^2 := Z_{\text{Gp}}^2((H, \circ), D; \sigma)$; its image $A(Z_\circ^2) \leq C^3$ is a subgroup.

Definition 1.12 (The solvability obstruction).

$$o_\nu(\beta) := [Q\beta] \in C^3/A(Z_\circ^2), \quad \beta \in Z_+^2.$$

The whole content of Theorem 1.3 is one formal identity.

Lemma 1.13 (Linchpin: the diagonal coboundary satisfies the coupling, [PROVED]). *Let $\theta: H \rightarrow D$, $\theta(0) = 0$, be arbitrary, set $\theta_1(h) := \nu_h^{-1}(\theta(h))$, and let $\beta := d_+\theta$ be the μ -twisted additive coboundary and $\bar{\tau} := d_\circ\theta_1$ the σ -twisted multiplicative coboundary. Then $P(T d_\circ\theta_1) = Q(d_+\theta)$ identically in θ .*

Proof. This is exactly $B_{\text{Sb}}^2 \subseteq Z_{\text{Sb}}^2$ restricted to component (c): the diagonal coboundary $(d_+\theta, d_\circ\theta_1)$ lies in Z_{Sb}^2 , so satisfies (1.3). Rathee–Yadav establish $B^2 \subseteq Z^2$ as a formal identity in θ and the triplet, requiring no brace to exist. The identity was additionally verified exhaustively by machine over all θ and all good triplets harvested from genuine braces, holding on 12/12, 32/32, 9/9, 192/192 instances for $(H, D) = (\mathbb{Z}/2, \mathbb{Z}/4), (\mathbb{Z}/2, (\mathbb{Z}/2)^2), (\mathbb{Z}/3, \mathbb{Z}/3), (\mathbb{Z}/4, \mathbb{Z}/4)$. $\square$

Theorem 1.14 (o_ν is a homomorphism and $\text{im } \varphi_+ = \ker o_\nu$, [PROVED]). *Fix the good triplet (ν, μ, σ) with D a trivial brace. Then:*

- (1) $o_\nu(d_+\theta) = 0$ for every θ ; hence o_ν descends to a map $o_\nu: H_+^2 \rightarrow C^3/A(Z_\circ^2)$, $[\beta] \mapsto [Q\beta]$.
- (2) o_ν is a group homomorphism, additive in $[\beta]$.
- (3) $\text{im } \varphi_+ = \ker o_\nu$. Equivalently: an additive class $[\beta]$ extends to a full skew-brace cocycle inducing (ν, μ, σ) iff $o_\nu([\beta]) = 0$.

Proof. (1) By Lemma 1.13, $Q(d_+\theta) = P(T d_\circ\theta_1) \in A(Z_\circ^2)$ since $d_\circ\theta_1 \in Z_\circ^2$; so o_ν kills B_+^2 and factors through $H_+^2 = Z_+^2/B_+^2$. (2) Q is $\mathbb{Z}$ -linear (each summand of (1.2) is additive in β because $\nu_{h_1}, \mu_{h_1 \circ h_3} \in \text{Aut}(D, +)$ and reindexing is additive) and $A(Z_\circ^2)$ is a subgroup. (3) If $[\beta] = \varphi_+[(\beta, \bar{\tau})]$ then $(\beta, \bar{\tau}) \in Z_{\text{Sb}}^2$, so by (1.3) $Q\beta = P(T\bar{\tau}) \in A(Z_\circ^2)$ and $o_\nu([\beta]) = 0$. Conversely, if $o_\nu([\beta]) = 0$ there is $\bar{\tau} \in Z_\circ^2$ with $P(T\bar{\tau}) = Q\beta$; then $(\beta, \bar{\tau})$ satisfies (a),(b),(c), so by RY’s characterisation $(\beta, \bar{\tau}) \in Z_{\text{Sb}}^2$ and $[\beta] = \varphi_+[(\beta, \bar{\tau})] \in \text{im } \varphi_+$. *This direction constructs the brace cocycle; it does not presuppose one — the circular step of the refuted argument is thereby avoided.* $\square$

The reformulation — “(b)+(c) is affine $A\tau = R_\nu(\beta)$; solvability = vanishing of $o_\nu(\beta)$; $\text{im } \varphi_+ = \ker o_\nu$ ” — is Rick’s Remark 4 from the referee report. The construction of the general- H operators (1.1)–(1.2), the descent via the linchpin identity, and the analysis below are the content added here.

Proposition 1.15 (Cross-validation, [COMPUTED]). *For $H = \mathbb{Z}/2$ with $D = \mathbb{Z}/4$ (all 4 triplets) and $D = (\mathbb{Z}/2)^2$ (all 16 triplets), and for $H = \mathbb{Z}/3$ with $D = \mathbb{Z}/3$, the abstractly computed $\ker o_\nu \subseteq H_+^2$ equals the $\text{Aut}(D)$ -saturated set of additive classes realised by genuine skew braces with the given triplet (independent regular-subgroup-of-holomorph enumeration). For $D = (\mathbb{Z}/2)^2$ the obstruction is genuinely nonzero for many triplets: it cuts $|H_+^2| = 4$ down to $|\ker o_\nu| = 2$, matched exactly — a stringent test of (1.1)–(1.2), the twist T , and the descent.*

This is the precise replacement for the retracted independence theorem. There is no product $R \times \Omega^0$: the realisable additive classes are $\ker o_\nu$, and $\ker o_\nu$ depends on ν through the $\nu_{h_1}(\beta(h_2, h_3))$ term of (1.2). The next section makes that dependence visible.

1.6 Explicit residue-dependence, and a torsor

$|G| = 8$: channel 1, linear (in the RY regime). Take $H = \mathbb{Z}/2$, $D = \mathbb{Z}/4$, μ trivial, $\sigma = \sigma_1$. Normalised cochains are single values $b = \beta(1, 1)$, $t = \bar{\tau}(1, 1) \in D$; here $\nu_{1 \circ 1} = \nu_0 = \text{id}$ so T acts trivially, $A(Z_\circ^2) = 2D^\sigma$, and o_ν evaluates to

$$o_\nu(b) = 0 \iff (1 + \nu)b \in 2D^\sigma.$$

With $\sigma = -1$, $D^\sigma = \{0, 2\}$ and $2D^\sigma = \{0\}$, so

$$\nu = 1 : 2b = 0 \iff b \in \{0, 2\} \iff [\beta] = 0; \quad \nu = -1 : 0 \in \{0\} \text{ always } \iff [\beta] \text{ free.}$$

So $\nu = 1$ *forces* $[\beta] = 0$ (the additive group must be $\mathbb{Z}/2 \times \mathbb{Z}/4$, not $\mathbb{Z}/8$), while $\nu = -1$ realises both classes. This is the decisive $|G| = 8$ coupling: the set of realisable $[\beta]$ genuinely depends on the residue, already inside the trivial-kernel (RY) regime. It is a corollary of Theorem 1.14 and matched by Proposition 1.15. Compare the refuted claim, which would have required $[\beta]$ to be free of ν .

$|G| = 16$: channel 2, three-way, [COMPUTED]. Take $D = \mathbb{Z}/8$ carrying the *non-trivial* brace $\lambda_x^D = 5^x$ (so $L = \{1, 5\} \subsetneq \text{Aut}(D)$), $H = \mathbb{Z}/2$. This is outside RY's trivial-kernel regime; the residue now enters a *second* channel, through λ_d on the additive complement, $\lambda_d(k) = k + \nu\sigma(d) - d$. Writing $b \in \mathbb{Z}/8$ for the additive 2-cocycle, every realising extension satisfies the **three-way lock**

$$(\lambda_d^D - 1)b = 2(\nu\sigma(d) - d) \quad \forall d, \quad \text{at } d = 1 : \quad \boxed{4b \equiv 2(\nu\sigma - 1) \pmod{8}}. \quad (1.4)$$

Verified on *all* 24 genuine order-16 braces with this kernel: 8/8 on $\mathbb{Z}/16$ ($[\beta] \neq 0$) and 16/16 on $\mathbb{Z}/2 \times \mathbb{Z}/8$ ($[\beta] = 0$). The residue and circle action enter (1.4) *only through the product* $\nu\sigma \pmod{4}$, which is then tied to the additive class:

$$[\beta] = 0 \text{ (} b \text{ even)} \iff \nu\sigma \equiv 1 \pmod{4}.$$

This is why on $\mathbb{Z}/2 \times \mathbb{Z}/8$ the residue $\{3, 7\}$ co-occurs with $[\beta] = 0$ *only* when $\sigma \in \{3, 7\}$, never with $\sigma = \text{id}$. The coupling is genuinely three-way $([\nu], \sigma, [\beta])$, not a two-way product.

What the G_0/G_1 corner really shows. The old draft's centrepiece was a pair of genuine order-16 braces G_0, G_1 on $(G, +) = \mathbb{Z}/2 \times \mathbb{Z}/8$ with $D \cong \mathbb{Z}/8$, $L = \{1, 5\}$, sharing residue $[\nu_1] = \{3, 7\}$ (with $\sigma = \cdot 3$) and $[\beta] = 0$, yet with $[\Omega] = 0$ for G_0 and $[\Omega] \neq 0$ for G_1 . This is a real computation and it says something true: $[\Omega]$ *is not a function of* $[\nu]$. But that is a statement about a *single fibre* $[\nu] = \{3, 7\}$ — it does *not* say the residue and the obstruction are independent. Indeed, an enumeration of all 176 labelled braces in the $\mathbb{Z}/16 \cup \mathbb{Z}/2 \times \mathbb{Z}/8$ family shows *no* value of σ realises the full product $[\nu] \times [\Omega]$: at $\sigma = \cdot 3$ the residue $[\nu] = L$ is not admissible at all, and at $\sigma \in \{\text{id}, \cdot 5\}$ the pair $([\nu] = \{3, 7\}, [\Omega] = 0)$ is missing. Reading “ $[\Omega]$ not determined by $[\nu]$ in one fibre” as “ $[\nu] \perp [\Omega]$ ” was the error. The fibrewise non-determination survives as a [COMPUTED] fact; the product does not.

The torsor: a non-trivial-kernel effect. In the trivial-kernel regime o_ν is *linear* (Theorem 1.14(2)), so $\text{im } \varphi_+ = \ker o_\nu$ is a *based* subgroup: it always contains 0, i.e. the split additive class $[\beta] = 0$ is realisable with any prescribed residue. The no-basepoint phenomenon is strictly a *non-trivial-kernel* effect, visible in (1.4): the right-hand side $2(\nu\sigma - 1)$ is **constant in** b . Thus the obstruction is *affine*, $4b = c$ with $c = 2(\nu\sigma - 1)$, and the realisable set of b is a *coset* of $\ker(\times 4) = \{0, 2, 4, 6\}$:

$$\nu\sigma \equiv 1 \pmod{4} : c = 0, b \in \{0, 2, 4, 6\} \ni 0 \Rightarrow [\beta] = 0 \text{ available;} \quad \nu\sigma \equiv 3 \pmod{4} : c = 4, b \in \{1, 3, 5, 7\} \not\ni 0,$$

a torsor with no basepoint: no split-additive brace exists with that (ν, σ) ; $[\beta] \neq 0$ is forced. This is the honest form of the “internal-replacement” statement the refuted product theorem was reaching for:

Changing the internal relabelling $[\nu]$ does not leave the available decomposition obstructions untouched; it moves them, through the affine offset $c = 2(\nu\sigma - 1)$. When the internal datum $\nu\sigma$ leaves the socle-like subgroup ($\nu\sigma \not\equiv 1$), the realisable locus $\text{im } \varphi_+$ is a torsor over $\ker o_\nu$ that misses the origin.

A general- H , general-kernel proof that $\text{im } \varphi_+$ is an o_ν -affine torsor (with the offset identified intrinsically as the class of the λ_d -complement cochain) is the natural next target; here it is [COMPUTED] on the $|G| = 16$ family and **open** in general.

1.7 Honest ledger, and Ferri’s Remark 3.20

Statement	Grade
$[\Omega] = [(\beta, \bar{\tau})] \in H_{\text{Sb}}^2(H, D)$ is the Rathee–Yadav class; φ_+ descends; $[\Omega] = 0 \iff$ bicrossed (trivial-kernel SH)	[PEER-REVIEWED]
Product map $\Phi = (\varphi_\circ, \varphi_+)$ non-injective (W_4)	[PEER-REVIEWED]
$[\Omega]$ is a bilinear class, strictly finer than both group Schreier classes; base- ζ localised; heap-transport refuted	[PROVED]
Residue invariance ($[\nu_h] = \nu_h L$ section-independent; torsor T_ν)	[PROVED]
Orthogonality fails at the cochain level (not a gauge-orbit product)	[PROVED]
$o_\nu: H_+^2 \rightarrow C^3/A(Z_\circ^2)$ well-defined group hom killing B_+^2 ; $\text{im } \varphi_+ = \ker o_\nu$ (trivial kernel, general H)	[PROVED]
$\ker o_\nu = \text{im } \varphi_+$ cross-validated vs genuine enumeration ($H = \mathbb{Z}/2$: $D = \mathbb{Z}/4, (\mathbb{Z}/2)^2$; $H = \mathbb{Z}/3$)	[COMPUTED]
$ G = 8$: $o_\nu(b) = 0 \iff (1 + \nu)b \in 2D^\sigma$; $\nu = 1$ forces $[\beta] = 0$ (channel 1, linear)	[PROVED]
$ G = 16$ three-way lock $4b \equiv 2(\nu\sigma - 1) \pmod{8}$; $\nu, \sigma, [\beta]$ locked through $\nu\sigma$ (channel 2, non-trivial kernel)	[COMPUTED]
Non-trivial kernel $\Rightarrow o_\nu$ affine $\Rightarrow \text{im } \varphi_+$ a torsor, no basepoint when $\nu\sigma \not\equiv 1$	[COMPUTED]
General- H /general-kernel affine-torsor theorem	open
φ_+ descent and W_4 non-injectivity decide certificates	[LEAN-VERIFIED]

Three honesty notes are load-bearing and belong on the record.

- *The independence claim is withdrawn.* A 2026-09-26 draft asserted a third theorem — $[\nu]$ an orthogonal direct factor, $\text{Ext} \cong R \times \Omega^0$. It is **refuted** (Rick, 2026-09-26; reproduced 2026-09-29; registry *cross-independence-nu-omega* demoted proved $\rightarrow$ computed, *residue-independent-obstruction-group* and *residue-surjectivity-beta-nonzero* $\rightarrow$ speculative). The error was a circular reduction: “read $\circ$ off from $+$ and λ ” presupposes the brace whose existence is in question, and the *homogeneous* coupling being ν -free does not make *inhomogeneous* solvability ν -free. What replaces it is the residue-*dependent* o_ν of §1.5.

- *No socle-necessity claim.* An earlier draft of the identification asserted separation “requires $D \not\subseteq \text{Soc}(G)$.” On referee review (Rick) this was found false and is *deliberately omitted*; it is not needed for any result here.
- *The general kernel is native.* The identification of $[\Omega]$ with a published cohomology (Rathee–Yadav) is proved only for the trivial-kernel standing hypothesis. Housing the general non-trivial-kernel obstruction in a skew-brace H^2 where ν is *varying* cocycle data is native to this programme and not yet reduced to any published theory; the $|G| = 16$ analysis and the torsor are [COMPUTED], and we do not claim otherwise.

Relation to Ferri’s Remark 3.20. Ferri [1] asks, in the language of quiver skew braces, for a group-theoretic reformulation of when an ideal is *prunable* — when the quiver skew brace decomposes off that ideal. Theorem 1.7 supplies exactly this: prunability of D is the vanishing of the single skew-brace class $[\Omega]$, equivalently the existence of a sub-skew-brace (star) complement; and Theorem 1.2 explains *why* no purely group-theoretic (single-operation) criterion can suffice — the obstruction is bilinear, and lives in the coupling that no forgetful functor to a group H^2 can retain. The base-localisation of Theorem 1.9 is the groupoid refinement: prunability is decided on one star and transported. Theorem 1.3 adds the parametric picture: as the internal residue varies, prunability is governed by the residue-dependent homomorphism o_ν , whose kernel — the realisable additive classes — moves with the residue.

What this feeds. For the grant, the chapter closes the skew-brace / Yang–Baxter instance of the decomposition programme with a clean separation of concerns: an *existence* axis ($[\omega]$, Chapter ??) and a strictly finer *decomposition* axis ($[\Omega]$, this chapter), the latter a bilinear invariant. The relation to the internal relabelling residue is *coupling*, not independence: the residue enters the solvability of the extension through the homomorphism o_ν , and the honest form of “internal replacement” is that re-indexing the internal action *moves* which decomposition obstructions are available — linearly in the trivial-kernel regime, affinely (with a possible loss of basepoint) once the kernel is non-trivial. The general- H /general-kernel affine-torsor theorem is the isolated open target for the next proof session.

Bibliography

- [1] D. Ferri, *On quiver skew braces, their ideals and products*, arXiv:2605.11903, 2026.
- [2] N. Rathee, M. K. Yadav, *Skew brace extensions, second cohomology and complements*, arXiv:2601.12371, 2026.